

SRI SIVASUBRAMANIYA NADAR COLLEGE OF ENGINEERING, KALAVAKKAM
(An Autonomous Institution, Affiliated to Anna University, Chennai)
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
III Year CSE (V Semester)
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UCS3513 SYSTEM DESIGN LABORATORY

Lab Exercise 4 - Design and simulate Horizontal Database Sharding using MongoDB

Objective

To deploy a MongoDB sharded cluster using Docker Compose, configure horizontal database sharding, distribute data across multiple shards, and analyze the scalability and query performance of a distributed database system.

Problem Statement

A university's Student Information System stores a large number of student records. As the volume of data grows, storing all records in a single MongoDB server leads to increased storage requirements and reduced query performance.

To improve scalability and load distribution, deploy a MongoDB sharded cluster using Docker Compose. Configure horizontal sharding for the **College** database, distribute the **Student** collection across multiple shards, and verify data distribution using MongoDB sharding commands.

Tasks

- Configure the Config Server, Shard Servers, and mongos Query Router.
- Add the shard servers to the MongoDB cluster.
- Enable sharding for the **College** database.
- Create and shard the **Student** collection using an appropriate shard key.
- Display the shard distribution using MongoDB commands.
- Execute search queries through mongos.
- Analyze how documents are distributed across the shards.
- Compare the advantages of a sharded database over a single MongoDB instance.

Analysis

1. Explain the role of:
 - Config Server
 - Shard Server
 - mongos Router
2. Justify the selection of RollNo as the shard key.
3. Explain what happens when additional shards are added.
4. Compare a standalone MongoDB server with a MongoDB sharded cluster.

Expected Learning Outcomes

After completing this experiment, students will be able to:

- Install and configure Docker Desktop.
- Create multi-container applications using Docker Compose.
- Deploy a MongoDB sharded cluster.
- Configure replica sets and shard servers.
- Understand the architecture of MongoDB sharding.
- Configure horizontal sharding using an appropriate shard key.
- Analyze data distribution and query routing.
- Explain the scalability benefits of distributed databases.